



Chapter 5: Results & Analysis

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5.1 Overview

This chapter presents the experimental results obtained from the instruction fine-tuning experiments described in Chapter 4. The purpose of this chapter is to analyze how different batch scheduling strategies affect optimization performance, generalization behavior, phase-wise training dynamics, and generation quality.

The evaluated strategies include random batching, semantic grouped batching, two-phase curriculum scheduling, and length-based curriculum scheduling. The main comparison is performed across Dolly 1k, 3k, and 5k subsets using multi-seed runs with seeds 13, 21, and 42. The optimization metrics include mean evaluation loss, standard deviation of evaluation loss, mean training loss, generalization gap, and phase-wise evaluation loss change. Generation quality is evaluated using ROUGE and BERTScore on the 5k experiments where generation evaluation artifacts are available.

The canonical source for optimization results is the aggregated master summary table. Generation metrics are taken from the aggregated generation summary files for the 5k batching and length-curriculum experiments. Since some generation aggregates conflict with overlapping per-seed JSON files, generation results are interpreted cautiously.

Overall, the results show that random batching remains a strong baseline. Semantic grouped batching does not consistently outperform random batching. Two-phase curriculum ordering shows some weak directional effects, especially for Random → Grouped compared with Grouped → Random, but the differences are small. Length-based curriculum, especially Hard → Easy ordering, provides the clearest positive signal, although the same-budget improvement remains small.

5.2 Optimization Results

This section reports the optimization results across the main experiment blocks. Lower evaluation loss indicates better optimization performance. Each reported value is averaged across three seeds: 13, 21, and 42.

5.2.1 Results on Dolly 1k Short Training

The first controlled experiment block compares random batching, grouped batching, grouped-to-random curriculum, and random-to-grouped curriculum on the Dolly 1k subset using short training. The results are shown in Table 5.1.

Table 5.1: Dolly 1k Short Training Comparison

Method	Experiment ID	N	Seeds	Mean Eval Loss	Std Eval Loss
Random	exp_012_random_only_multiseed	3	13,21,42	9.616399129231771	0.0001284643317367
Grouped	exp_013_grouped_only_multiseed	3	13,21,42	9.61652692159017	0.0002104496546978
Grouped → Random	exp_014_grouped_to_random_multiseed	3	13,21,42	9.616591453552246	0.0000126159253648
Random → Grouped	exp_015_random_to_grouped_multiseed	3	13,21,42	9.616386731465658	0.000090394709832

In this block, Random → Grouped achieves the lowest mean evaluation loss, while Grouped → Random gives the highest mean evaluation loss. However, the difference between the best and worst methods is only `0.000204722086588`, which is

extremely small. Therefore, this result suggests a possible ordering effect, but it does not support a strong performance claim.

The important observation is that Random → Grouped slightly improves over Random by only `0.000012397766113`. This is too small to treat as a practically meaningful improvement. The result is better interpreted as a weak signal that the order of batching may influence training dynamics.

5.2.2 Results on Dolly 1k Longer Training

The second 1k experiment block uses longer training while keeping the same four strategy types. The results are shown in Table 5.2.

Table 5.2: Dolly 1k Longer Training Comparison

Method	Experiment ID	N	Seeds	Mean Eval Loss	Std Eval Loss
Random	<code>exp_018_random_only_multiseed_long</code>	3	13,21,42	9.611108779907227	0.000186715695
Grouped	<code>exp_019_grouped_only_multiseed_long</code>	3	13,21,42	9.611552556355795	0.000108124687
Grouped → Random	<code>exp_020_grouped_to_random_multiseed_long</code>	3	13,21,42	9.611759503682455	0.000210961151
Random → Grouped	<code>exp_021_random_to_grouped_multiseed_long</code>	3	13,21,42	9.611469268798828	0.00024478796

In the longer 1k setting, Random achieves the lowest mean evaluation loss. Grouped → Random again performs worst among the four strategies. Random → Grouped is better than Grouped → Random, but it is still worse than Random.

This result weakens the claim that Random → Grouped is consistently better overall. Instead, it suggests that Random → Grouped may be better than the reverse ordering, but random batching remains difficult to outperform.

5.2.3 Results on Dolly 3k

The 3k experiments test whether the same patterns persist at a larger dataset scale. The results are shown in Table 5.3.

Table 5.3: Dolly 3k Training Comparison

Method	Experiment ID	N	Seeds	Mean Eval Loss	Std Eval Loss
Random	<code>exp_022_random_only_multiseed_3k</code>	3	13,21,42	11.423059145609537	0.002296378511
Grouped	<code>exp_023_grouped_only_multiseed_3k</code>	3	13,21,42	11.425052642822266	0.000682492412
Grouped → Random	<code>exp_024_grouped_to_random_multiseed_3k</code>	3	13,21,42	11.424866358439127	0.001265824868
Random → Grouped	<code>exp_025_random_to_grouped_multiseed_3k</code>	3	13,21,42	11.425841649373373	0.00068428796

At 3k, Random again achieves the lowest mean evaluation loss. Random → Grouped, which was strongest in the 1k short block, performs worst in the 3k block. This shows that the Random → Grouped advantage is not consistent across dataset scales.

The difference between the best and worst methods is `0.002782503763836`, which is still small. This supports the interpretation that batching strategy affects training only weakly under the tested conditions.

5.2.4 Results on Dolly 5k

The 5k experiment block is the strongest main comparison because it uses the largest dataset scale among the core random/grouped/curriculum experiments. The results are shown in Table 5.4.

Table 5.4: Dolly 5k Main Training Comparison

Method	Experiment ID	N	Seeds	Mean Eval Loss	Std Eval Loss
Random	exp_026_random_only_multiseed_5k	3	13,21,42	11.341803868611654	0.004030287838
Grouped	exp_027_grouped_only_multiseed_5k	3	13,21,42	11.346779187520346	0.002508390861
Grouped → Random	exp_028_grouped_to_random_multiseed_5k	3	13,21,42	11.344831148783365	0.000999571402
Random → Grouped	exp_029_random_to_grouped_multiseed_5k	3	13,21,42	11.343379974365234	0.002895475059

In the 5k main comparison, Random achieves the lowest mean evaluation loss. Grouped batching performs worst, with a mean evaluation loss of 11.346779187520346, compared with 11.341803868611654 for Random. The difference between Grouped and Random is 0.004975318908692.

Although this is the largest pure Grouped-versus-Random gap among the directly comparable blocks, it remains small in absolute terms and close to the observed standard deviation for Random. The result therefore supports the conclusion that semantic grouping does not provide a reliable optimization advantage over random batching.

5.3 Random vs Semantic Grouping

A central hypothesis of this thesis was that semantically grouped mini-batches might improve fine-tuning by producing more coherent training updates. The canonical results do not support this hypothesis.

Across all four directly comparable blocks, pure Grouped batching performs worse than Random:

Block	Grouped Minus Random Eval Loss	Interpretation
1k Short	0.000127792358399	Grouped is slightly worse
1k Long	0.000443776448568	Grouped is worse
3k	0.001993497212729	Grouped is worse
5k Main	0.004975318908692	Grouped is worse

Since lower evaluation loss is better, all positive differences indicate that Grouped performs worse than Random. This is an important negative result. It shows that increasing semantic coherence within mini-batches does not automatically improve instruction fine-tuning.

One possible explanation is that instruction-tuning datasets are heterogeneous. Random batches expose the model to diverse task types within training updates, while semantic grouping may reduce within-batch diversity. In this setting, diversity may be more useful than local semantic coherence.

The safest conclusion is that semantic grouped batching does not produce a consistent or meaningful improvement over random batching. Random batching remains a robust baseline and should not be treated as a weak default.

5.4 Curriculum Ordering Analysis

The two-phase curriculum experiments compare two orderings:

1. **Random → Grouped**, interpreted as exploration followed by semantic refinement.
2. **Grouped → Random**, interpreted as early structure followed by later randomization.

The direct comparison between these two strategies is shown in Table 5.5.

Table 5.5: Curriculum Ordering Comparison

Block	Random → Grouped Mean Eval Loss	Grouped → Random Mean Eval Loss	Better Ordering	Difference
1k Short	9.616386731465658	9.616591453552246	Random → Grouped	-0.000204722086588
1k Long	9.611469268798828	9.611759503682455	Random →	-0.000290234883627

Block	Random → Grouped Mean Eval Loss	Grouped → Random Mean Eval Loss	Better Ordering	Difference
			Grouped	
3k	11.425841649373373	11.424866358439127	Grouped → Random	0.000975290934246
5k Main	11.343379974365234	11.344831148783365	Random → Grouped	-0.001451174418131

Random → Grouped performs better than Grouped → Random in three of the four blocks: 1k short, 1k long, and 5k main. However, the 3k block reverses this pattern, where Grouped → Random performs better.

This suggests a weak directional tendency in favor of Random → Grouped, but the evidence is not strong enough to claim that Random → Grouped is consistently superior. The margins are also small in all cases.

The results are compatible with an exploration–refinement interpretation. Random batching in the first phase may expose the model to broader task diversity, while grouped batching in the second phase may provide more focused updates. However, because Random → Grouped does not consistently beat the Random baseline, the result should be interpreted as a training-dynamics signal rather than a clear performance improvement.

5.5 Phase-2 Evaluation Loss Analysis

The phase-wise evaluation loss change measures how much the evaluation loss changes during the second phase of training. A negative value indicates that evaluation loss decreased during Phase 2. Therefore, more negative values indicate stronger improvement during the second phase.

Table 5.6: Phase-2 Evaluation Loss Change

Block	Method	Mean Phase2 ΔEval	Interpretation
1k Short	Random	-0.0013834635416666667	Strongest phase-2 improvement in block
1k Short	Grouped	-0.0013230641682942708	Improvement, weaker than Random
1k Short	Grouped → Random	-0.001311937967936198	Weakest improvement in block
1k Short	Random → Grouped	-0.0013634363810221355	Improvement, close to Random
1k Long	Random	-0.004041353861490886	Strongest phase-2 improvement in block
1k Long	Grouped	-0.0038159688313802085	Improvement, weaker than Random
1k Long	Grouped → Random	-0.0037781397501627603	Weakest improvement in block
1k Long	Random → Grouped	-0.003803253173828125	Improvement, weaker than Random
3k	Random	-0.03380934397379557	Strongest phase-2 improvement in block
3k	Grouped	-0.032903035481770836	Improvement, weaker than Random
3k	Grouped → Random	-0.03327242533365885	Second strongest improvement
3k	Random → Grouped	-0.032347679138183594	Weakest improvement in block
5k Main	Random	-0.07824929555257161	Strongest phase-2 improvement in block
5k Main	Grouped	-0.07535489400227864	Improvement, weaker than Random
5k Main	Grouped → Random	-0.07722759246826172	Improvement, close to Random
5k Main	Random → Grouped	-0.07583808898925781	Improvement, weaker than Random
5k Length	Easy → Hard Length	-0.0795135498046875	Strong phase-2 improvement
5k Length	Hard → Easy Length	-0.08008988698323567	Slightly stronger than Easy → Hard
Extended	Hard → Easy Longer Training	-0.2848030726114909	Much larger, but longer-budget result

All phase-2 delta values are negative. This means that the second training phase reduces evaluation loss in every experiment block and for every strategy. However, within the same-budget comparisons, the differences between methods remain small.

The Random strategy often has the strongest phase-2 improvement among the random/grouped comparison blocks. This further supports the strength of random batching as a baseline. The much larger phase-2 improvement in the extended hard-to-easy experiment is important, but it must be interpreted separately because it uses a longer training budget.

5.6 Length-Based Curriculum Results

The length-based curriculum experiments compare two ordering strategies on the Dolly 5k dataset:

1. **Easy → Hard Length**, where shorter examples are presented earlier.
2. **Hard → Easy Length**, where longer examples are presented earlier.

The standard-budget comparison is shown in Table 5.7.

Table 5.7: 5k Length Curriculum Comparison

Method	Experiment ID	N	Seeds	Mean Eval Loss	Std Eval Loss
Easy → Hard Length	exp_030_easy_to_hard_length_multiseed_5k	3	13,21,42	11.338643074035645	0.0012300206
Hard → Easy Length	exp_031_hard_to_easy_length_multiseed_5k	3	13,21,42	11.337978998819986	0.0009772446

Hard → Easy Length achieves a lower mean evaluation loss than Easy → Hard Length. The exact difference is `0.000664075215659` in favor of Hard → Easy. Hard → Easy also shows a slightly stronger phase-2 improvement, with a mean phase-2 delta of `-0.08008988698323567` compared with `-0.0795135498046875` for Easy → Hard.

Although this result favors Hard → Easy, the same-budget difference is very small. Therefore, it should be treated as a promising signal rather than a strong conclusion.

The extended hard-to-easy experiment is shown in Table 5.8.

Table 5.8: Extended Hard-to-Easy Training Result

Method	Experiment ID	N	Seeds	Mean Eval Loss	Std Eval
Hard → Easy Length	exp_031_hard_to_easy_length_multiseed_5k	3	13,21,42	11.337978998819986	0.00097
Hard → Easy Length Longer Training	exp_032_hard_to_easy_length_longer_training_5k	3	13,21,42	10.985996882120768	0.00812

The longer-training Hard → Easy experiment achieves the lowest mean evaluation loss across the reported experiments: `10.985996882120768`. This is substantially lower than the standard-budget Hard → Easy result of `11.337978998819986`.

However, this comparison is not same-budget. The improvement cannot be attributed only to the Hard → Easy ordering because the longer experiment also uses more training. The correct interpretation is that Hard → Easy length-based scheduling combined with longer training gives the strongest observed result, but it should not be presented as a clean same-budget strategy win.

A further caveat is that the length curriculum implementation uses a score based on tokenized sequence length. Since processed inputs are padded to a fixed maximum length, the input-length component may be largely constant. As a result, the ordering is likely driven mainly by non-padded label length rather than full example difficulty. This limits strong claims about "difficulty" in the curriculum design.

5.7 Generation Quality Results

Generation quality was evaluated for the 5k batching and length-curriculum experiments using ROUGE and BERTScore. The combined generation results are shown in Table 5.9.

Table 5.9: Generation Metrics on 5k Experiments

Method	Mean ROUGE-1	Mean ROUGE-2	Mean ROUGE-L	Mean ROUGE-Lsum	Mean BERTScore F1
Random	0.1332991221165225	0.05172597342614166	0.12528074807592213	0.1255176984312918	0.835802110791206
Grouped	0.13515864447757928	0.05267632000153163	0.12741708266264154	0.12762222208271706	0.83652363093694
Grouped → Random	0.13573552053686477	0.05290762532282947	0.12741556756984396	0.1279101682552873	0.83624990793065
Random → Grouped	0.13429778290121783	0.05139917328310489	0.12623164340206786	0.1263183377758188	0.83633620341618
Easy → Hard Length	0.1349535952911857	0.051761622149622204	0.12602883537238033	0.12640203541955244	0.836186007102331
Hard → Easy Length	0.13569431685742261	0.052930931439159314	0.12673431381120112	0.12715204647000258	0.836213118275006

The best-performing method differs depending on the metric. Grouped → Random obtains the highest ROUGE-1 and ROUGE-Lsum. Hard → Easy Length obtains the highest ROUGE-2. Grouped obtains the highest ROUGE-L and BERTScore F1.

The metric spread is small across all methods:

Metric	Spread Across Methods
ROUGE-1	0.00243639842034227
ROUGE-2	0.001531758156054424
ROUGE-L	0.00213633458671941
ROUGE-Lsum	0.0023924698239955
BERTScore F1	0.0007215201457343

These small spreads show that the generation results do not support strong claims of substantial output-quality improvement from any single batching strategy. Although grouped and curriculum methods sometimes score slightly higher than Random on individual metrics, no method dominates all metrics.

Therefore, the safest conclusion is that batching strategy has minimal measurable effect on generation quality under the current automated evaluation setup.

5.8 Stability and Seed Variance

Multi-seed evaluation is important because many observed differences are extremely small. The standard deviation values show that seed-level variation is often comparable to method-level differences.

Table 5.10: Seed Variance and Stability Summary

Block	Method	Std Eval Loss	Comment
1k Short	Grouped	0.00021044965469782815	Full method spread is smaller than this standard deviation
1k Long	Random → Grouped	0.0002447879672933447	Method differences are still very small in absolute terms
3k	Random	0.0022963785111432176	Method spread is only modestly larger than Random std
5k Main	Random	0.004030287838901957	Method spread is close to the largest standard deviation
5k Length	Easy → Hard Length	0.0012300206513966253	Same-budget method spread is smaller than both curriculum std values
Extended	Hard → Easy Longer Training	0.008129767881074827	Larger std, but still far smaller than the mean improvement over standard-budget Hard → Easy

This analysis limits the strength of the conclusions. When the difference between two methods is similar to the standard deviation across seeds, it is not safe to claim a strong or statistically significant improvement. Since no formal statistical significance test was conducted, the results should be interpreted as empirical trends rather than definitive proof.

The strongest exception is the extended Hard → Easy result, where the mean improvement over standard-budget Hard → Easy is much larger than the reported standard deviation. However, this is still not a same-budget comparison.

5.9 Summary of Findings

This chapter analyzed the results of batch scheduling strategies for instruction fine-tuning. The main findings are as follows.

First, random batching remains a strong and reliable baseline. It achieves the best mean evaluation loss in the 1k longer training, 3k, and 5k main comparison blocks. In the 1k short block, Random → Grouped slightly outperforms Random, but only by a very small margin.

Second, semantic grouped batching does not provide a consistent optimization advantage. Pure Grouped batching performs worse than Random in all four directly comparable blocks. This suggests that increasing within-batch semantic coherence is not sufficient to improve instruction fine-tuning performance under the tested setup.

Third, curriculum ordering affects training dynamics, but the effect is weak. Random → Grouped performs better than Grouped → Random in three of the four comparable blocks, which supports a possible exploration-refinement interpretation. However, the effect is not consistent across all scales and does not consistently outperform Random.

Fourth, phase 2 improves evaluation loss in all experiments. Every reported phase-2 delta is negative, indicating that the second phase contributes additional optimization. Nevertheless, within same-budget blocks, differences between strategies remain small.

Fifth, length-based curriculum provides the clearest positive signal. Hard → Easy Length performs slightly better than Easy → Hard Length under the same 5k budget. The extended Hard → Easy experiment achieves the best overall mean evaluation loss, but this result must be interpreted as a longer-training result rather than a clean same-budget comparison.

Finally, generation metrics show only small differences across methods. No method dominates all ROUGE and BERTScore metrics, and the observed metric spreads are narrow. Therefore, the generation evaluation does not support strong claims that batching strategy substantially improves output quality.

Overall, the results suggest that batch scheduling can influence training dynamics, but its effect on final performance is limited. Random batching remains difficult to outperform, semantic grouping alone is not sufficient, and curriculum-based ordering—especially hard-to-easy length scheduling—appears more promising but still requires careful interpretation.

The results show that random batching remains a strong baseline, semantic grouping does not provide a consistent optimization advantage, curriculum ordering has some effect but remains small, and length-based hard-to-easy scheduling provides the clearest positive signal. However, most same-budget differences are small and often comparable to seed variation, so the findings should be interpreted conservatively.